Multi-backend deep learning framework for multimodal microscopy

Inés Varona-Peña[1], Sara Cruz-Adrados[1], Rosa-María Menchón-Lara[2], and Biagio Mandracchia[1]

[1]*Universidad de Valladolid, Valladolid, Spain.*

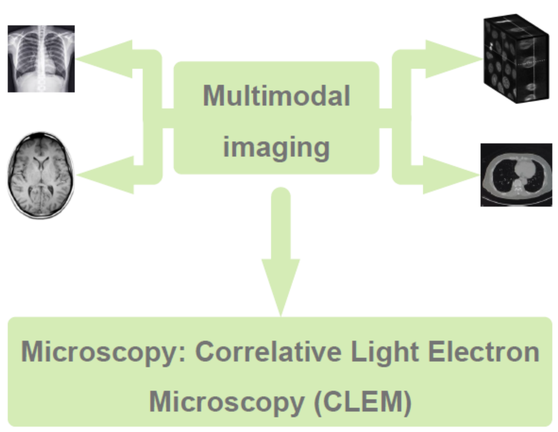
[2]*Universidad Politécnica de Cartagena member of European University of Technology EUT+, Spain.*

ABSTRACT

We present **bioMONAI**, an open-source deep learning framework for multimodal microscopy that extends the MONAI ecosystem with microscopy-oriented tools. The framework provides native support for heterogeneous microscopy data, domain-aware preprocessing, specialized loss functions, and advanced visualization utilities. Through **Keras 3 integration**, bioMONAI enables multi-backend compatibility across PyTorch, TensorFlow, and JAX, facilitating seamless transitions between research prototyping and large-scale deployment. Rather than introducing new learning algorithms, bioMONAI focuses on reducing implementation complexity, improving reproducibility, and enabling modular, end-to-end AI workflows in advanced microscopy settings.

INTRODUCTION

Multimodal imaging is the integration of two or more imaging modalities, which provides a more comprehensive understanding of biological processes



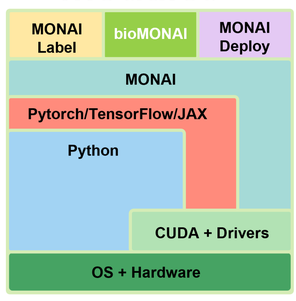
AI methods improve segmentation, classification, and restoration, but integrating heterogeneous microscopy data into reproducible pipelines remains challenging.

TensorFlow PyTorch MONAI JAX

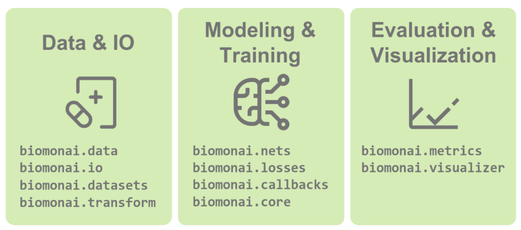
bioMONAI
Deep learning for biomedical imaging

IMPLEMENTATION

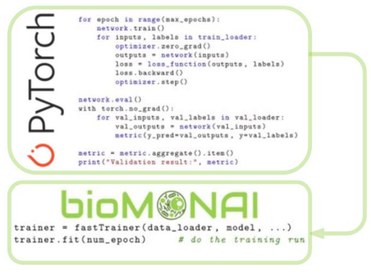
Modules



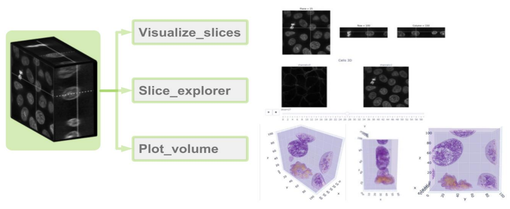
Low-code platform



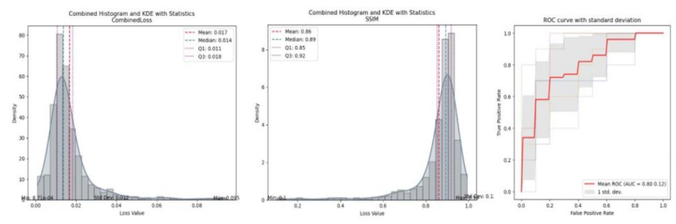
Low-code platform



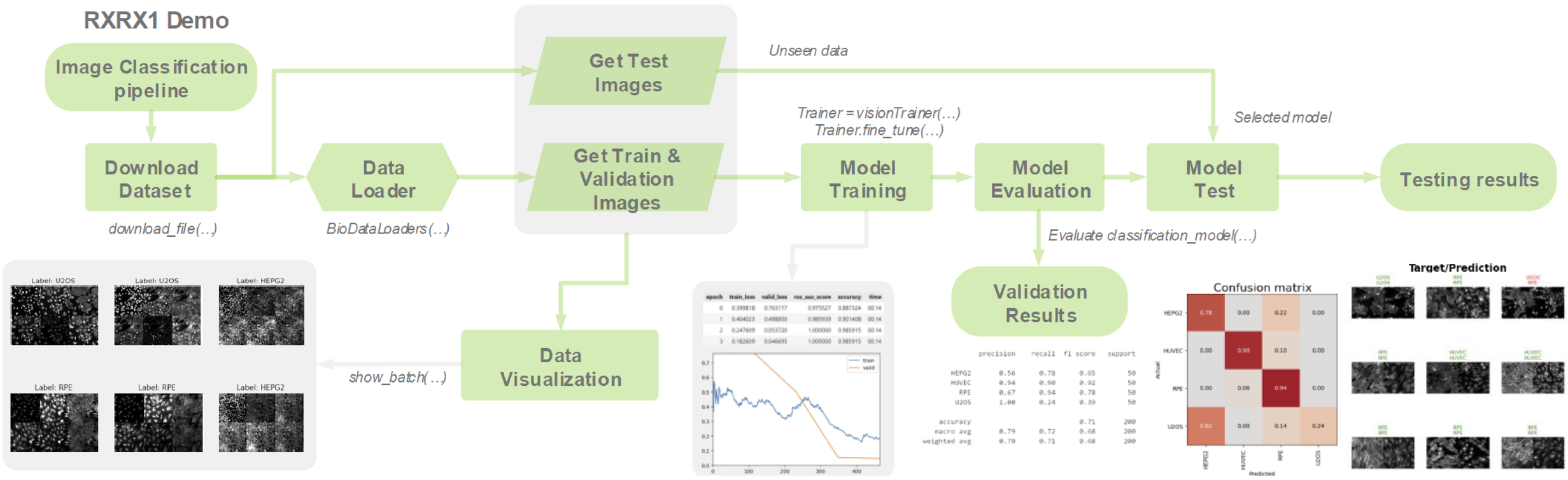
Visualization Tools



Evaluation Utilities



WORKFLOW EXAMPLE: CLASSIFICATION



CONCLUSIONS

bioMONAI is an open-source framework that extends the MONAI ecosystem to address the specific demands of multimodal biomedical imaging, particularly in advanced microscopy workflows and CLEM.

By combining microscopy-aware tools with Keras 3–enabled multi-backend interoperability, bioMONAI supports seamless execution across PyTorch, TensorFlow, and JAX, facilitating scalable and reproducible AI pipelines.

github.com/deepCLEM/bioMONAI

Available at: github.com/deepCLEM/bioMONAI

REFERENCES

[1] M. J. Cardoso *et al.*, "MONAI: An open-source framework for deep learning in healthcare", *arXiv [cs.LG]*, 2022.

[2] J. Bischof *et al.*, "Multimodal bioimaging across disciplines and scales: challenges, opportunities and breaking down barriers", *npj Imaging*, vol. 2, núm. 1, 2024.

[3] P. de Boer, J. P. Hoogenboom, y B. N. G. Giepmans, "Correlated light and electron microscopy: ultrastructure lights up!", *Nat. Methods*, vol. 12, núm. 6, pp. 503–513, 2015.

bioMONAI's docs!

bioMONAI's Github!

deepCLEM's docs!

This work was supported by the Spanish '*Ministerio de Ciencia, Innovación y Universidades*'-'*Agencia Estatal de Investigación*' (MICIU-AEI) and FEDER, UE, under grant PID2022-142166NA-I00. B. M. and R.M. M.-L. are supported by the Spanish MICIU-AEI '*Ramón y Cajal*' program under grants RYC2021-032084-I and RYC2023-045078-I, respectively.

MINISTERIO DE CIENCIA, INNOVACIÓN Y UNIVERSIDADES

Cofinanciado por la Unión Europea

AGENCIA ESTATAL DE INVESTIGACIÓN